KachaBench: Prior Work and Our Contribution

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Summary

KachaBench is a simulation study of **when, how and why a humanoid robot fails while walking on monsoon village roads** in Bangladesh: wet clay that sinks and sucks at the feet, deep wheel ruts, and potholes whose depth is hidden by muddy water. A Unitree G1 humanoid walks a procedurally generated earthen road in MuJoCo. The mud is modelled from soil mechanics using measured Bangladeshi clay properties. Instead of ranking many methods, the study tests a few controllers in depth across controlled mud conditions and records the cause of every fall.

Prior work

Legged robots have been trained for rough, rigid and granular (sandy) ground, and one quadruped study measured real mud, but **no work studies humanoids on cohesive wet clay with suction, hidden-depth water, or how and why they fail there.**

Work	What it did	What it does not cover
Lee et al., Science Robotics 2020	Quadruped (ANYmal) trained on rigid terrain in simulation; walked on mud, snow and water in the field	Mud never modelled; no analysis of why it fails there
Choi et al., Science Robotics 2023	Granular-media contact model for RL; quadruped running on beach sand	Sand only: no clay cohesion, no suction; quadruped
Godon et al., IEEE/ASME TMECH 2025	*Walking in Mud*: Go1 quadruped in real mud about 12 cm deep; modelling, control and experiments	Quadruped; one mud type; no humanoid, water or load

Work	What it did	What it does not cover
GenTe, arXiv 2025	Terrains generated from text and images with vision-language models; wading water and sand; bipedal robot	Generic scenes; sand-like soil; water has no hidden depth; no failure analysis
MILD, 2025	Discrete-element model of yielding ground for a large flat foot; biped tested on foam, grass and sand	Accuracy of contact on dry soft ground; no wet clay, suction or water
GM-Loco, arXiv 2026	Unitree G1 humanoid on granular media with a resistive-force contact model; sim and hardware on sand	The closest work: same robot, but dry sand only; no cohesion, suction, water, load or failure-cause study
Radosavovic et al., Science Robotics 2024	RL humanoid (Digit) walking in the real world on varied everyday ground	Rigid and mildly uneven ground; no soft or wet terrain
Cham & Redfern 2002; Gates et al. 2012	How people change their gait on slippery or unstable ground: shorter strides, flatter landing, higher foot lift	Human studies; never compared with robot policies

Soil mechanics for the mud model: Skempton (1951) footing bearing capacity on clay, Matlock (1970) lateral resistance, and measured Atterberg limits of Bangladeshi clays (Hasnat et al. 2020; Islam et al. 2004).

The gap

Prior work leaves four things open, and this study addresses each:

1. **Cohesive wet clay.** Soft-ground models are granular (sand). Monsoon clay behaves differently: it sinks under load, resists sideways and holds a lifting foot by suction.
2. **Hidden hazards.** Water fills potholes and ruts, so a camera or height scanner sees a flat surface and the robot learns the depth only by stepping in.
3. **Humanoids on wet ground.** Quadrupeds have been tested in mud; humanoids, with two feet and a high centre of mass, have only been tested on sand.

4. **Why failure happens.** Papers report success rates. None measures the cause of each fall (slip, sinking, suction, hidden water, balance), or how the mix of causes changes with conditions.

No work models South Asian rural roads or monsoon-season conditions for legged robots.

What we test

A Unitree G1 humanoid (35 kg, 1.32 m, 29 motors) must walk 8.3 m along a muddy, rutted village road in simulation without falling or leaving the road.

Research questions

1. **Breakdown:** at what soil wetness, mud depth, friction, suction and water level does walking break down, and how sharply?
2. **Mechanism:** what share of falls are slips, sinking, suction trapping a foot, hidden water, or balance loss, and how does the mix change with the mud?
3. **Training conditions:** does a controller trained on rigid ground, or on one season, hold up in another?
4. **Load (later):** does carrying a box in both hands, or a load on the head as station porters do, bring the breakdown earlier?
5. **Gait adaptation:** do trained controllers change their gait on worse mud the way people do on slippery ground?

What is varied, what is compared, what is measured

Part	Contents
Mud conditions (one at a time)	soil wetness (liquidity index), mud depth, surface friction, suction strength, water in potholes and ruts; each across its plausible range
Controllers	Unitree's pretrained rigid-ground policy; the scripted mud-walking gait alone; RL policies trained on the dry road, on early-monsoon mud and on mixed seasons; one with short-term memory; 3 training seeds each
Per episode	success; the trigger of each fall and how it fell (tipped or collapsed); step length, cadence, double support, foot lift, sinkage, sliding, cost of transport

Part	Contents
Statistics	30-50 test roads per condition; success with 95% confidence intervals

Our contributions

1. **A wet-clay mud model for humanoid feet**, built from soil mechanics (Skempton bearing capacity, Matlock lateral resistance, suction on lift-off) and parameterised by measured Bangladeshi clay properties. Soil wetness sets strength and suction together, so one physical quantity controls the mud. It runs fast enough for RL training on a 4-core CPU.
2. **Hidden-depth hazards:** potholes sized from the LGED rural-road guideline and rut puddles filled with opaque water, so perception sees only the surface.
3. **To our knowledge, the first study of humanoid walking on cohesive mud**, with robustness curves across the plausible range of each mud property rather than single assumed values.
4. **Failure-mechanism analysis:** every fall is attributed to a trigger (slip, sinking, suction, hidden water, balance loss), so results explain *why* controllers fail, not only how often.
5. **A rigid-to-mud transfer test** of a manufacturer's controller (Unitree's pretrained G1 policy) against policies trained on mud.
6. **A comparison with human gait adaptation:** whether learned controllers discover the same changes people make on slippery ground.
7. **An open, low-cost setup:** MuJoCo on a CPU, with code, terrain generator and evaluation scripts released, usable by groups without GPUs.

Status and limitations

The simulation, mud model, task, controllers and analysis tools are built and tested. The first 50M-step training run started on 4 November 2026; **there are no results yet.**

Done	Not yet
Mud and water models, verified against their own equations	Validation against measured mud (published foot-force data or a field test)

Done	Not yet
G1 on the muddy road, scripted mud gait, training pipeline	A trained policy that crosses the road
Failure-cause recording, gait measures, robustness sweeps	Robustness curves and failure analysis
Unitree baseline adapter	Running Unitree's policy (needs its model file on the PC)
	Load carrying (hand box, head load)

Limitations to state plainly:

- **Simulation only.** No real robot; findings describe the simulated physics.
- **Season values are assumptions.** No field data on village-road moisture was found; the sweeps cover the whole plausible range to limit how much results depend on them.
- **Reduced-order mud.** No viscosity or rate effects; holes refill when a foot leaves; the sole is treated as flat on the mud.
- **Heuristic failure triggers.** Thresholds were set by inspecting traces and need a sensitivity check.
- **Compute.** About 19 hours per training run on the 4-core PC, so 4 controllers x 3 seeds is about 9-10 days.

Questions a supervisor may ask

Question	Short answer
How is this different from GM-Loco, which also puts a G1 on soft ground?	GM-Loco models dry sand (granular). We model cohesive wet clay with sinkage, sideways resistance and suction, plus water-hidden potholes, and we study failure causes rather than one controller's success.

Question	Short answer
How do you know the mud is realistic?	Every force comes from published soil mechanics and measured Bangladeshi clay limits, and the code reproduces its equations. It is not yet validated against real mud; a published Go1 mud dataset or a simple field test (weighted plate plus fall cone) is the next step.
Why simulation and not a real robot?	No humanoid is available, and mud would risk damaging one. Simulation also allows controlled sweeps of one property at a time, which field tests cannot.
Why a humanoid and not a quadruped?	Two feet and a high centre of mass make mud far more dangerous, humanoids are the open problem, and only humanoids can carry loads the way people on these roads do. The quadruped (Go1) was used first to validate the mud model.
Why not a benchmark with many methods?	A benchmark needs compute for many methods. A robustness study tests a few controllers in depth and explains failures, which fits a 4-core CPU and gives more insight.
What if the trained policy never learns to walk?	The scripted gait and Unitree's policy still give curves and failure causes. Training also uses a scripted base gait and a fading balance assist to make learning easier.
Are the season settings measured?	No; they are reasoned assumptions. The study sweeps each property over its plausible range, so conclusions do not rest on one value.
What is publishable here?	Stage 1 with failure analysis suits a robotics workshop; adding mud validation and load carrying moves it toward a conference or journal such as IEEE RA-L.

Question	Short answer
How long will it take?	About 9-10 days of training on the PC for the core controllers, plus a few days of sweeps and analysis.

References

The literature check is not exhaustive; a full Google Scholar search is needed before writing.

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